Bus and Train Ticket Booking Interface Design

1. Introduction

The Bus and Train Ticket Booking Interface is a web-based application designed to make travel booking simple, convenient, and user-friendly. It allows users to enter their source and destination, select travel dates, provide passenger details, view available buses and trains, compare fares, and select seats. The interface uses forms, tables, and responsive booking cards to display information clearly and improve the overall user experience.

2. Features

- Source and Destination: Users can enter their starting point and destination.
- Travel Date Selection: Users can choose their preferred departure date and return date.
- Passenger Details: The interface allows users to enter passenger information and select the number of travellers.
- Available Services: Displays available bus and train services with departure times, arrival times, journey duration, and service details.
- Fare Information: Shows ticket prices to help users compare travel options.
- Seat Selection: Provides a seat-selection layout to view and choose available seats.
- Booking Forms: Collects journey and passenger details in an organized manner.
- Tables: Displays schedules, service details, and fare information in a structured format.
- Responsive Booking Cards: Presents travel options in cards that adapt to different screen sizes.
- Booking Summary: Displays the selected journey, passenger details, seats, and total fare.

3. Applications

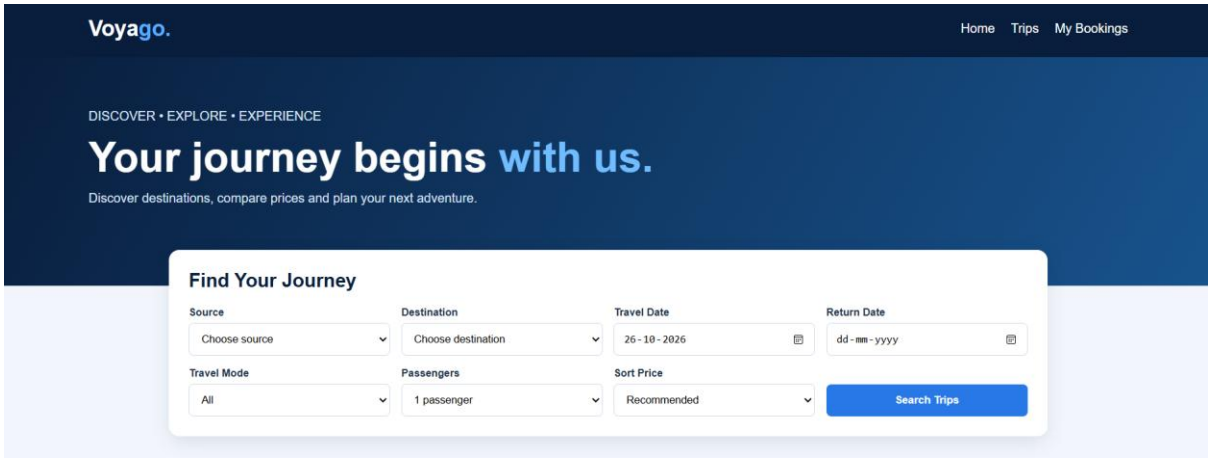
- Online bus and train ticket reservation websites.
- Public transportation booking systems.
- Travel agencies and tour-planning platforms.
- Transport company websites for displaying schedules and fares.

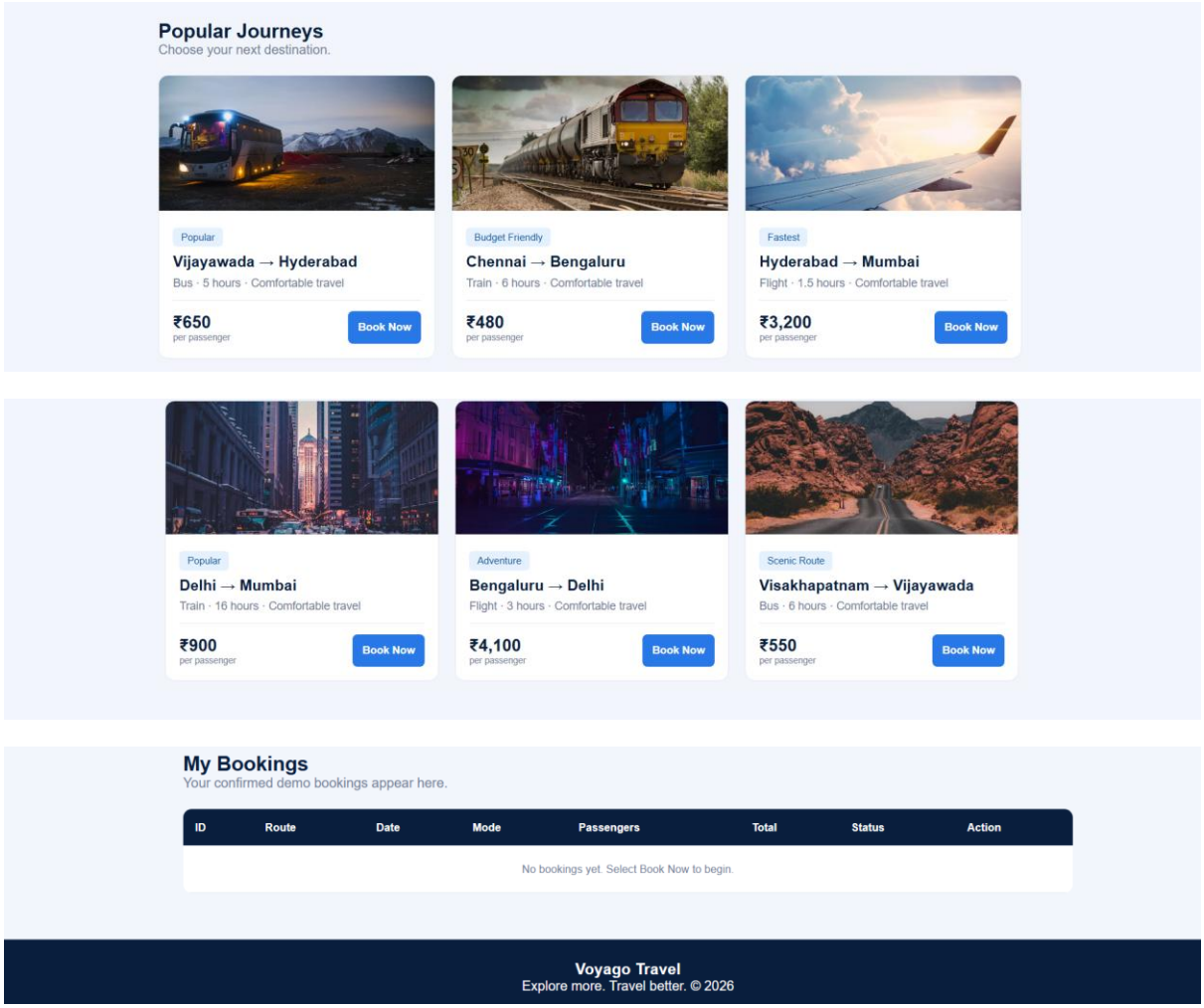
- Educational projects demonstrating web development and responsive interface design.
- Travel-planning applications that help users compare different transportation options.

4. Future Scope

- Live Availability: Integrate real-time bus and train schedules and seat availability.
- Online Payments: Add secure payment gateway integration.
- User Login: Provide user registration, login, and profile management.
- E-Ticket Generation: Generate downloadable tickets with booking reference numbers.
- Booking History: Allow users to view previous and upcoming bookings.
- Cancellation and Refunds: Add ticket cancellation, rescheduling, and refund tracking.
- Advanced Search: Provide filters based on fare, departure time, journey duration, and seat type.
- Multilingual Support: Support multiple languages for a wider range of users.
- Mobile Optimization: Improve the interface for smartphones and tablets.

5. Output Screenshots





6. Source Code

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width,initial-scale=1">

<title>Voyago Travel Booking</title>

<style>

*{box-sizing:border-box;margin:0;padding:0}

body{font-family:Arial,sans-serif;background:#f2f6fc;color:#20324a}

header{background:#091e3d;color:white;padding:20px 7%;display:flex;justify-content:space-between;align-items:center;flex-wrap:wrap;gap:15px}

```
.logo{font-size:27px;font-weight:bold;color:white}

.logo span{color:#58adff}

nav{display:flex;gap:20px;flex-wrap:wrap}

nav a{color:white;text-decoration:none;transition:.3s}

nav a:hover{color:#58adff}

.hero{padding:65px 7% 100px;background:linear-gradient(120deg,#091e3d,#18538d);color:white}

.hero h1{font-size:clamp(32px,5vw,52px);margin:12px 0}

.hero span{color:#71bdff}

.hero p{color:#dbe7f5}

.search{max-width:1100px;margin:-45px auto 0;background:white;padding:25px;border-radius:13px;box-shadow:0 8px 25px #091e3d18}

.search h2{margin-bottom:18px;color:#091e3d}

form,.fields{display:grid;grid-template-columns:repeat(4,minmax(0,1fr));gap:14px}

label{font-size:13px;font-weight:bold;display:block;margin-bottom:6px}

input,select{width:100%;padding:11px;border:1px solid #ccd8e8;border-radius:6px;background:white}

input:focus,select:focus{outline:2px solid #58adff}

button{cursor:pointer;border:0;border-radius:6px;padding:12px;background:#2878e8;color:white;font-weight:bold;transition:.25s}

button:hover{background:#091e3d;transform:translateY(-2px)}

main{max-width:1100px;margin:55px auto;padding:0 20px}

.heading{display:flex;justify-content:space-between;gap:12px;flex-wrap:wrap;margin-bottom:22px}

.heading h2{color:#091e3d}

.heading p{color:#718096}

.cards{display:grid;grid-template-columns:repeat(3,minmax(0,1fr));gap:20px}
```

```
.card{background:white;border:1px solid #dfe7f1;border-radius:12px;overflow:hidden;transition:.3s}

.card:hover{transform:translateY(-7px);box-shadow:0 12px 28px #091e3d20}

.card img{width:100%;height:165px;object-fit:cover}

.tag{display:inline-block;background:#e5f1ff;color:#1558a0;padding:5px 10px;border-radius:5px;font-size:12px;margin-bottom:8px}

.content{padding:17px}

.content h3{color:#091e3d;margin-bottom:7px}

.content p{color:#718096;font-size:14px;margin-bottom:12px}

.bottom{display:flex;justify-content:space-between;align-items:center;gap:8px;border-top:1px solid #e5eaf2;padding-top:13px}

.price{font-size:21px;font-weight:bold;color:#091e3d}

.price small{display:block;font-size:11px;color:#718096;font-weight:normal}

.table-wrap{overflow-x:auto;background:white;border-radius:10px}

table{width:100%;min-width:600px;border-collapse:collapse}

th,td{padding:14px;text-align:left;border-bottom:1px solid #e2e9f2;font-size:13px}

th{background:#091e3d;color:white}

tr:hover td{background:#edf5ff}

.status{color:#137443;background:#dcfce7;padding:4px 8px;border-radius:12px}

.cancel{background:#ffe7e7;color:#a12222;padding:6px}

.empty{text-align:center;padding:20px;color:#718096}

footer{background:#091e3d;color:white;text-align:center;padding:25px;margin-top:50px}

#toast{position:fixed;bottom:20px;right:20px;background:#091e3d;color:white;padding:15px;border-radius:8px;display:none;max-width:90%}

@media(max-width:800px){.cards{grid-template-columns:repeat(2,minmax(0,1fr))}form,.fields{grid-template-columns:repeat(2,minmax(0,1fr))}}

@media(max-width:520px){header{padding:18px 5%}nav{gap:12px}.hero{padding:45px 6% 80px}.cards{grid-template-columns:1fr}form,.fields{grid-template-columns:1fr}.search{margin:-35px 12px 0;padding:18px}main{padding:0 14px}}
```

```
</style>

</head>

<body>


<header>

<div class="logo">Voya<span>go.</span></div>

<nav><a href="#home">Home</a><a href="#trips">Trips</a><a href="#bookings">My Bookings</a></nav>

</header>


<section class="hero" id="home">

<p>DISCOVER • EXPLORE • EXPERIENCE</p>

<h1>Your journey begins <span>with us.</span></h1>

<p>Discover destinations, compare prices and plan your next adventure.</p>

</section>


<section class="search">

<h2>Find Your Journey</h2>

<form id="searchForm">

<div><label for="src">Source</label><select id="src" required><option value="">Choose source</option><option>Vijayawada</option><option>Hyderabad</option><option>Chennai</option><option>Bengaluru</option><option>Delhi</option><option>Mumbai</option><option>Visakhapatnam</option></select></div>

<div><label for="dest">Destination</label><select id="dest" required><option value="">Choose destination</option><option>Vijayawada</option><option>Hyderabad</option><option>Chennai</option><option>Bengaluru</option><option>Delhi</option><option>Mumbai</option><option>Visakhapatnam</option></select></div>

<div><label for="date">Travel Date</label><input id="date" type="date" required></div>

<div><label for="ret">Return Date</label><input id="ret" type="date"></div>
```

<label for="mode">Travel Mode</label><select id="mode"><option>All</option><option>Bus</option><option>Train</option><option>Flight</option></select>

<label for="people">Passengers</label><select id="people"><option value="1">1 passenger</option><option value="2">2 passengers</option><option value="3">3 passengers</option><option value="4">4 passengers</option><option value="5">5 passengers</option></select>

<label for="sort">Sort Price</label><select id="sort"><option value="default">Recommended</option><option value="low">Lowest first</option><option value="high">Highest first</option></select>

<button style="width:100%">Search Trips</button>

</form>

</section>

<main>

<section id="trips">

<h2>Popular Journeys</h2> Choose your next destination.

</section>

<section id="bookings" style="margin-top:55px">

<h2>My Bookings</h2> Your confirmed demo bookings appear here.

<table><thead><tr><th>ID</th><th>Route</th><th>Date</th><th>Mode</th><th>Passengers</th><th>Total</th><th>Status</th><th>Action</th></tr></thead><tbody id="rows"><tr><td colspan="8" class="empty">No bookings yet. Select Book Now to begin.</td></tr></tbody></table>

</section>

</main>

<footer><h3>Voyago Travel</h3> Explore more. Travel better. © 2026 </footer>

<script>

const trips=[

{a:"Vijayawada",b:"Hyderabad",mode:"Bus",price:650,time:"5 hours",tag:"Popular",img:"photo-1544620347-c4fd4a3d5957"},

{a:"Chennai",b:"Bengaluru",mode:"Train",price:480,time:"6 hours",tag:"Budget Friendly",img:"photo-1474487548417-781cb71495f3"},

{a:"Hyderabad",b:"Mumbai",mode:"Flight",price:3200,time:"1.5 hours",tag:"Fastest",img:"photo-1436491865332-7a61a109cc05"},

{a:"Delhi",b:"Mumbai",mode:"Train",price:900,time:"16 hours",tag:"Popular",img:"photo-1519501025264-65ba15a82390"},

{a:"Bengaluru",b:"Delhi",mode:"Flight",price:4100,time:"3 hours",tag:"Adventure",img:"photo-1519608487953-e999c86e7455"},

{a:"Visakhapatnam",b:"Vijayawada",mode:"Bus",price:550,time:"6 hours",tag:"Scenic Route",img:"photo-1500530855697-b586d89ba3ee"}

];

const $=id=>document.getElementById(id);

let bookings=[],nextId=1001,results=[...trips];

const today=new Date();

const localDate=new Date(today.getTime()-today.getTimezoneOffset()*60000).toISOString().slice(0,10);

$("date").min=localDate;

$("ret").min=localDate;

$("date").value=localDate;

```
$("date").addEventListener("change",()=>{
 $("ret").min=$("date").value;
 if($("ret").value && $("ret").value<$("date").value)$("ret").value="";
});

function notify(message){
 $("toast").textContent=message;
 $("toast").style.display="block";
 clearTimeout(notify.timer);
 notify.timer=setTimeout(()=>$("toast").style.display="none",2800);
}

function renderTrips(){
 const grid=$("cards");
 grid.innerHTML="";
 if(!results.length){
  grid.innerHTML='<div class="empty">No matching trips. Try another route or travel mode.</div>';
  return;
 }
 results.forEach(t=>{
  const original=trips.indexOf(t);
  const card=document.createElement("article");
  card.className="card";
  card.innerHTML=`
   <img src="https://images.unsplash.com/${t.img}?auto=format&fit=crop&w=800&q=80" alt="${t.mode} journey">
   <div class="content">
```

```
      <span class="tag">${t.tag}</span>
      <h3>${t.a} → ${t.b}</h3>
      <p>${t.mode} · ${t.time} · Comfortable travel</p>
      <div class="bottom">
       <div class="price">₹${t.price.toLocaleString("en-IN")}<small>per passenger</small></div>
       <button class="book" data-index="${original}">Book Now</button>
      </div>
     </div>`;
   grid.appendChild(card);
  });
}

$("searchForm").addEventListener("submit",e=>{
  e.preventDefault();
  const a=$("src").value,b=$("dest").value;
  if(a===b){notify("Source and destination must be different.");return;}
  if($("ret").value && $("ret").value<$("date").value){notify("Return date must follow departure date.");return;}
  results=trips.filter(t=>(!a||t.a===a)&&(!b||t.b===b)&&($("mode").value==="All"||t.mode===$("mode").value));
  if($("sort").value==="low")results.sort((x,y)=>x.price-y.price);
  if($("sort").value==="high")results.sort((x,y)=>y.price-x.price);
  renderTrips();
  $("trips").scrollIntoView({behavior:"smooth"});
});

$("sort").addEventListener("change",()=>{
  if($("sort").value==="low")results.sort((a,b)=>a.price-b.price);
```

```
 if($("sort").value==="high")results.sort((a,b)=>b.price-a.price);
 if($("sort").value==="default")results=trips.filter(t=>
  (!$("src").value||t.a===$("src").value)&&
  (!$("dest").value||t.b===$("dest").value)&&
  ($("mode").value==="All"||t.mode===$("mode").value));
 renderTrips();
});

$("cards").addEventListener("click",e=>{
 const btn=e.target.closest(".book");
 if(!btn)return;
 const t=trips[Number(btn.dataset.index)];
 if($("src").value&&$("src").value!==t.a){notify("Choose a trip matching your selected source.");return;}
 if($("dest").value&&$("dest").value!==t.b){notify("Choose a trip matching your destination.");return;}
 if($("mode").value!=="All"&&$("mode").value!==t.mode){notify("Choose a trip matching your travel mode.");return;}
 if(!$("date").value||$("date").value<localDate){notify("Please select a valid departure date.");$("searchForm").scrollIntoView({behavior:"smooth"});return;}
 const count=Number($("people").value);
 bookings.push({id:nextId++,route:t.a+" → "+t.b,date:$("date").value,mode:t.mode,count,total:t.price*count});
 renderBookings();
 notify("Demo booking confirmed! Booking ID: "+bookings[bookings.length-1].id);
 $("bookings").scrollIntoView({behavior:"smooth"});
});

function renderBookings(){
```

```
 const body=$("rows");
 body.innerHTML="";
 if(!bookings.length){
  body.innerHTML='<tr><td colspan="8" class="empty">No bookings yet.</td></tr>';
  return;
 }
 bookings.forEach(b=>{
  const row=document.createElement("tr");
  [b.id,b.route,b.date,b.mode,b.count,"₹"+b.total.toLocaleString("en-
IN")].forEach(value=>{
   const td=document.createElement("td");
   td.textContent=value;
   row.appendChild(td);
  });
  const status=document.createElement("td");
  status.innerHTML='<span class="status">Confirmed</span>';
  row.appendChild(status);
  const action=document.createElement("td");
  const cancel=document.createElement("button");
  cancel.className="cancel";
  cancel.textContent="Cancel";
  cancel.onclick=()=>{
   bookings=bookings.filter(item=>item.id!==b.id);
   renderBookings();
   notify("Demo booking cancelled.");
  };
  action.appendChild(cancel);
  row.appendChild(action);
```

```
    body.appendChild(row);
  });
}

renderTrips();
</script>
</body>
</html>
```

7. Conclusion

The Bus and Train Ticket Booking Interface provides a simple and organized platform for planning journeys and exploring travel options. By combining source and destination fields, travel dates, passenger forms, service listings, fare tables, and seat-selection layouts, the interface makes the booking process easier to understand. In the future, features such as real-time availability, online payments, e-ticket generation, and booking management can make the system more efficient and practical.